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Floral nectar production and nectary structure of a bee-pollinated shrub from Neotropical savanna

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Keywords

Anemopaegma album; floral nectar dynamics; nectary ultrastructure; plant–pollinator interaction.

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ABSTRACT

Biotic pollination is critical for tropical ecosystem functioning, and nectar plays an essential role as it represents the main trophic resource for pollinators. Nevertheless, little is known about the mechanisms that underlie its production, which is essential for understanding the basis of nectar-mediated interactions in ecological and evolutionary approaches. Therefore, this study explores the relationship between the nectar secretion pattern and nectary functional changes in *Anemopaegma album*, a bee-pollinated species. We analysed the pattern of nectar production under field conditions and investigated floral nectary structural changes in two different developmental stages using light, transmission and scanning electron microscopy. We measured 30.95 ± 23.02 µl (mean ± SD, n = 30) of nectar accumulated inside the nectar chamber (29.26 ± 3.48% sucrose equivalents) at the moment of flower opening. Nectar removal did not influence the pattern of floral nectar production in terms of volume or total sugar but reduced the concentration of the nectar produced during the first 24 h of anthesis. The nectary consisted of an epidermis, a nectary parenchyma and a subnectary parenchyma supplied only by phloem. Starch grains decreased in size and abundance from the subnectary parenchyma toward the epidermis. We observed the degradation of starch grains and incorporation of amyloplasts into vacuoles at the pre-anthesis stage as well as the transformation of amyloplasts into elaioplasts during anthesis. Nectar secretion was continuous during the *A. album* flower life span, which was related to the functional features of its floral nectary, especially the presence of starch stored in the parenchyma.

INTRODUCTION

Changes in plant–pollinator interactions associated with global changes (Burkle *et al.* 2013) are especially concerning in tropical communities because 94% of all angiosperm species in this region are pollinated by animals (Ollerton *et al.* 2011). Although nectar is the principal energetic resource for pollinators (Schmidt-Lebuhn *et al.* 2007), little is known about the relationship between nectar production and nectary traits, especially at the cellular level (Rivera 2000; Nepi 2007; Galetto 2009). Information on the relationships among nectar and nectary traits thus facilitates our understanding of the outcome of plant–pollinator interactions (*e.g.* Herrera 2000; Sun *et al.* 2013) and plant evolution (*e.g.* Galen 1996).

Nectar traits could be essential for establishing a reciprocal positive association, such as mutualism between plants and pollinators (Thomson & Wilson 2008). For example, bees tend to prefer more concentrated nectar in small volumes rather than the same energetic content in a larger volume of dilute nectar (Cnaani *et al.* 2006). This trend appears to be true for Bignoniaceae species, and Galetto (2009) found a higher nectar concentration and a lower nectar volume per flower in species that were pollinated by bees when compared with species pollinated by hummingbirds.

Because nectar is the primary resource available to the pollinators of most Bignoniaceae species and considering that pollinators can modulate nectar traits over evolutionary time, we expected to find an association between the nectar secretion pattern and structural and functional features of the floral nectary. Considering this scenario, we selected as our study model the shrub *Anemopaegma album* (Bignoniaceae) from Neotropical savanna, a plant that shows slight dichogamy and has nectar as the primary attractor for bee pollinators (*Guimarães et al.* 2015). Here, we focus first on the nectar secretion dynamics in terms of volume, concentration and total sugar per flower and second on the structural and functional aspects of nectar secretory cells. Our aim was to understand the mechanisms related to offering and maintaining suitable nectar for pollinators throughout the flower life span. More specifically, we addressed the following questions: (i) how does the pattern of floral nectar production vary throughout anthesis; (ii) is there any effect of nectar removal on the pattern of floral nectar production throughout anthesis; and (iii) can changes in the structural and ultrastructural features of nectary cells during floral development explain the nectar secretion pattern?

MATERIAL AND METHODS

Study area and plants

This study was conducted at the State Park of Grão Mogol in northern Minas Gerais state, Brazil. This region is characterised

by mountains, the vegetation occurs in oligotrophic and acidic soils, is subject to daily fluctuations in temperature, wind exposure and water restrictions, and vegetation is typically xeromorphic (Giulietti *et al.* 1997). Within Bignoniaceae, the genus *Anemopaegma*, pollinated by large- and medium-sized solitary bees (Gentry 1974), comprises approximately 40 species, distributed predominantly in forests, but with a group of species growing in a typical seasonally dry climate, as is the case for our focal plant species, *Anemopaegma album* Mart. ex DC (Lohmann & Taylor 2014).

Floral nectar secretion pattern

General procedure

We investigated the nectar secretion dynamics and total energetic content secreted over the floral life span during the flowering peak of January 2010. Although we observed that the flowers remained attached to inflorescences for 4 days, they were only functional during the first 2 days (Guimarães *et al.* 2015). Our experiments were performed until the beginning of the third day of anthesis. For this, we previously protected inflorescences containing pre-anthesis buds using bridal veil bags (voile fabric with a 0.1–0.3-mm mesh) to avoid nectar depletion by floral visitors during evaluation of the nectar production per flower, as suggested by Corbet (2003). We collected the nectar using calibrated glass syringes, and measured the sugar concentration with a digital refractometer, expressed as a percentage (w/w). Nectar pH was determined with pH paper sets. We also used nectar volumes and concentrations to estimate total milligrams sugar produced per flower using the exponential regression proposed in Galetto & Bernardello (2005). We used three floral nectar descriptors to evaluate the pattern of nectar dynamics throughout anthesis: (i) nectar volume; (ii) nectar concentration; and (iii) total sugar content per flower.

Because floral nectar production during anthesis could change after successive bee visits (*e.g.* the nectar removal effect), we evaluated the dynamics of nectar production in three steps, using flowers without manipulation and flowers with experimental successive nectar removals (simulating bee visits). An illustrative scheme of Steps 1, 2 and 3 is described in Appendix S1.

Step 1: nectar dynamics in unmanipulated flowers

We used unmanipulated flowers to verify the total nectar volume and sugar content produced per flower throughout the first 48 h of anthesis. A total of 111 recently opened flowers (N_{total}), from 30 plants, were placed in five subgroups ($N_1 = 30$, $N_2 = 18$, $N_3 = 18$, $N_4 = 24$, $N_5 = 21$). In this experiment, the flowers of each subgroup were sampled just once and discarded immediately after nectar removal. We performed the nectar sampling at intervals of 12 h, starting at 07:00 h (moment of flower opening) on the first day of anthesis ($N_1 = 30$ flowers) and finishing at 07:00 h on the third day of anthesis ($N_5 = 21$ flowers), for a total of 48 h of monitoring.

Step 2: nectar dynamics in flowers with nectar removal

We simulated the potential bee effect of nectar removal during the flower life span in a set of 36 flowers from 23 plants. We sampled and re-sampled the floral nectar in each flower at intervals of 12 h, starting at 07:00 h on the first day of anthesis ($N = 36$ flowers) and finishing at 19:00 h on the second day ($N = 17$ flowers), for a total of 36 h of monitoring. Each flower was evaluated four times, with the exception of some flowers that were evaluated just twice or three times due to the shorter flower life span or early corolla abscission (probably due to flower manipulation during nectar withdrawal). We did not consider the measurements taken after 48 h of anthesis because only two of the labelled flowers remained with the corolla attached to the flower.

Step 3: comparison of nectar dynamics between unmanipulated flowers and flowers with successive nectar removals

To determine the effect of nectar removal, we compared two distinct sets of data, one from Step 1 and the other from Step 2. The first set of data corresponds to the nectar sampled at 24 h of anthesis ($t_{(3)}$ from Step 1, $N_3 = 18$ flowers). The second set of data corresponds to the sum of nectar sampled in the three successive removals from each manipulated flower (nectar at $t_{(1)}$, $t_{(2)}$, $t_{(3)}$ from Step 2, $N = 29$ flowers). In this latter case, we have a single accumulated nectar value per flower under the effect of nectar removal. In both cases, we used nectar data from 0 to 24 h after anthesis, the period during which we had most of the flowers attached to the inflorescences.

Statistical analysis

In all steps, the nectar descriptors (volume, concentrations and total sugar per flower) were used as dependent variables.

In Step 1, we modelled the dynamics of nectar production by adjusting the variation of accumulated floral nectar (volume, concentration and sugar per flower) during 48 h of flower life span, using two distinct models: (i) a simple linear model and (ii) a quadratic linear model, similar to models proposed in Galetto & Bernardello (2005). These models describe two alternative hypotheses: a linear increase in nectar production throughout anthesis (first model), and an initial increase in nectar production with subsequent evaporation or re-absorption of accumulated nectar throughout anthesis (second model). To evaluate the importance of each model (simple and quadratic) we selected the best one by comparing AIC (Akaike's information criterion) values, indicated by the lowest AIC values. If the difference between the model's AIC values (ΔAIC) varied from 0 to 2, we have similar support or plausibility for both models. In this particular case, we adopt the simplest model, with fewer estimated parameters as suggested by Burnham & Anderson (2002).

In Step 2 we modelled the relationship between the floral nectar removed every 12 h (raw data of nectar volume, concentration and total sugar per flower) and the flower life span using a general linear mixed model (LMM). In this case, we used flower life span as fixed variable and re-sampled flowers as random variables.

In Step 3 we performed a *t*-test to compare the accumulated nectar between two treatments: unmanipulated flowers (control) and flowers with successive nectar removal (simulating bee visits). Since, in the latter treatment, we summed the volumes of successive nectar removals in each flower, we did not apply any random statement in this analysis as each nectar measurement from the same flower is not a replication.

For all analyses we considered each flower as an independent observation due to the high variation in nectar descriptors among flowers from the same plant, which was similar to the variation among flowers from different plants (data not shown); in addition to the fact that we had some plants with just one flower sampled. The homogeneity of variances was evaluated by looking at dispersion graphs of residuals. We obtained all statistics from untransformed data based on the dispersion graphs and visual inspection of nectar descriptors. Statistical analyses were carried out in R version 3.0.1 (R Development Core Team 2009) with the basic and lme4 packages.

Anatomical and ultrastructural analyses of floral nectary

For anatomical studies, we fixed nectaries from pre-anthesis buds (50–60 mm in length) and functional flowers on the first day of anthesis in FAA 50 (formaldehyde, acetic acid, 50% ethanol 1:1:18 v/v; Johansen 1940). The nectary samples were subsequently subjected to a gradual dehydration in an ethanol series and embedded in hydroxyethyl-methacrylate. We cut transverse and longitudinal sections (6–8 μm) with a rotary microtome and stained them with 0.05% toluidine blue (pH 4.3; O'Brien *et al.* 1964) for general histology. Histochemical tests were performed on hand-cut sections of fixed samples, including: Lugol reagent for starch grains, ruthenium red for pectins and mucilage (Johansen 1940), PAS technique for total polysaccharides (Jensen 1962), Sudan IV for total lipids, and ferric chloride for phenolic substances (Johansen 1940).

For scanning electron microscopy (SEM), we fixed samples in 2.5% glutaraldehyde (0.1 M phosphate buffer, pH 7.2), dehydrated them in an ethanol series, dried them to critical point, and subsequently sputter-coated them with approximately 10 nm gold (Robards 1978).

For transmission electron microscopy (TEM), we fixed the samples in glutaraldehyde (2.5% with 0.1 M phosphate buffer, pH 7.3) and left them overnight at 4 °C. We then post-fixed samples with 1% osmium tetroxide (OsO_4) in the same buffer for 2 h at room temperature, dehydrated them in a graded series of acetone solutions, and embedded them in Araldite resin. We stained ultrathin sections with uranyl acetate and lead citrate (Reynolds 1963).

RESULTS

The nectar secretion pattern of *Anemopaegma album* flowers

In general, floral nectar production in almost 80% of the flowers of *A. album* started at pre-anthesis and was quite viscous, making it difficult to withdraw with a syringe in some cases. We recorded no nectar in 13% of flowers at the moment of opening, and zero values were used for all volume calculations. The nectar secretion in recently opened natural flowers had on average a nectar volume of 30.95 ± 23.02 μl (mean ± SD; n = 30 flowers) inside the nectar chamber, with a sugar concentration of 29.26 ± 3.48%, mass/mass (mean ± SD; n = 30 flowers), which corresponded to 12.18 ± 7.14 mg sugar per flower (mean ± SD; n = 30 flowers). For nectar from recently opened flowers, we also detected a nectar pH of *ca.* 6–7. The floral nectar secretion pattern is described below in three complementary steps.

Step 1: nectar dynamics in unmanipulated flowers

Variations in nectar accumulated per flower, volume and total sugar during anthesis were best described with the quadratic linear model (Fig. 1A and C). This model described a gradual increase in nectar volume accumulated during the first day, showing a peak in nectar production at 24 h of anthesis, and a continuous decrease of volume toward the end of the 48 h of anthesis (Fig. 1A; $\Delta AIC_{(Linear\text{-}Quadratic)} = 8$; and statistics of quadratic model selected: $F_{(2,108)} = 4.89$, $P = 0.009$). The concentration of floral nectar decreased slightly throughout anthesis (Fig. 1B; $\Delta AIC_{(Linear\text{-}Quadratic)} = 2$; statistics of quadratic model selected: $F_{(1,93)} = 6.02$, $P = 0.016$) and total milligrams sugar produced per flower exhibited a similar pattern for nectar volume during anthesis (Fig. 1C; $\Delta AIC_{(Linear\text{-}Quadratic)} = 5$; statistics of quadratic model selected: $F_{(2,92)} = 3.79$, $P = 0.026$).

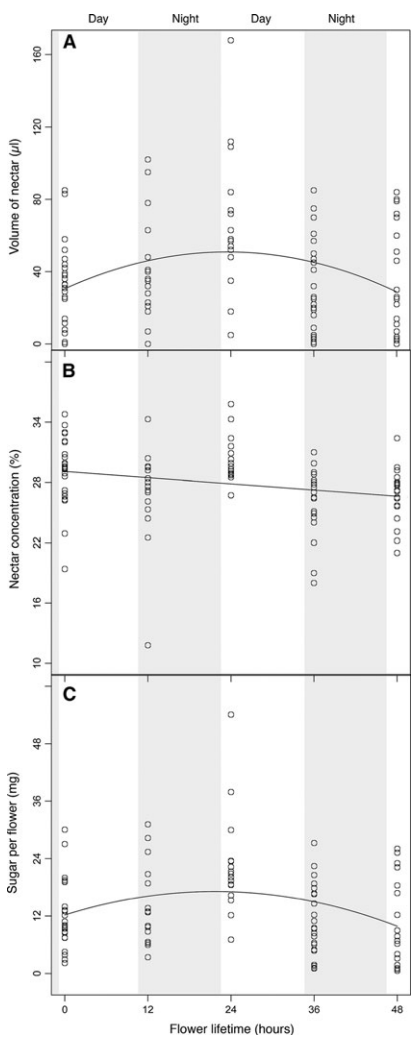


Fig. 1. Accumulated floral nectar recorded in unmanipulated flowers of *Anemopaegma album* during anthesis. A: Volume of nectar accumulated per flower (μl). B: Nectar concentration (percentage sucrose equivalents). C: Total sugar produced per flower (mg). Each dot represents a single flower, and the day and night periods are highlighted in white and grey, respectively. ($n_{0\ h} = 30$ flowers, $n_{12\ h} = 18$ flowers, $n_{24\ h} = 18$ flowers, $n_{36\ h} = 24$ flowers, $n_{48\ h} = 21$ flowers from 30 plants; for more details, see Appendix S1).

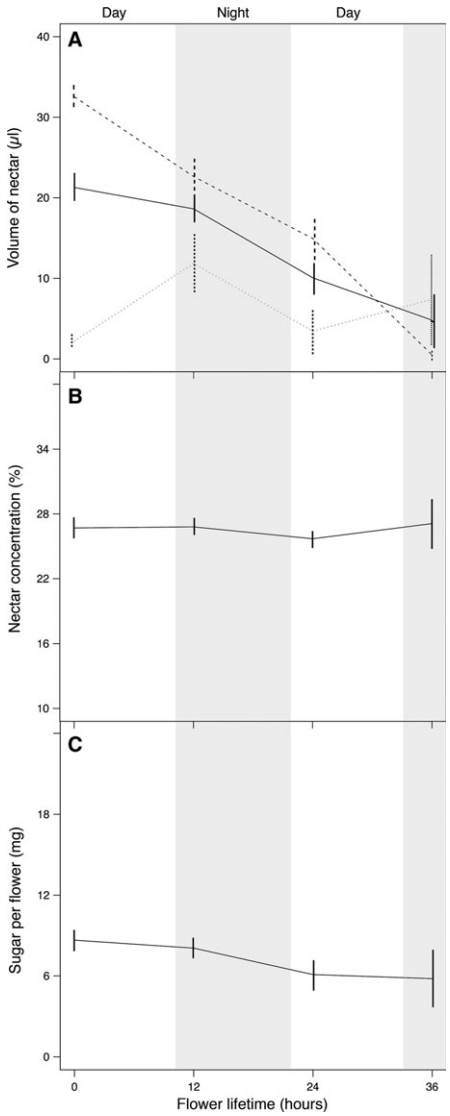


Fig. 2. Absolute values of floral nectar secretion re-sampled successively at 12-h intervals (mean ± SE) from *Anemopaegma album* flowers. A: Volume of nectar (µl). The solid line represents the general effect of lifetime on the volume of nectar secreted per 12 h, including all sampled flowers. The dashed line represents the volume of nectar secreted every 12 h in flowers with high initial nectar production. The dotted line represents the volume of nectar produced every 12 h in flowers with low or zero initial nectar production. B: Nectar concentration (percentage sucrose equivalents). C: Total sugar produced per flower at each interval of 12 h (mg). The day and night periods are highlighted with white and grey, respectively. (n = 36 flowers from 23 plants; for more details, see Appendix S1).

Step 2: nectar dynamics in flowers with successive nectar removals
For this group of flowers (simulating bee visits) we observed a gradual decrease in nectar production during the first 36 h of anthesis ($F_{(1,68)} = 7.74$, $P = 0.007$; Fig. 2A, solid line). Production practically ceased at the end of the second day in flowers that started anthesis with around 30 µl nectar stored in the nectar chamber (dashed line in Fig. 2A). Additionally, flowers that started anthesis with zero or very few microliters of nectar accumulated inside the nectar chamber showed a different pattern. We recorded small quantities of nectar produced during the first and second days of anthesis, the last being even smaller (dotted line in Fig. 2A), contributing, on average, to total nectar production (solid line in Fig. 2A). We did not observe significant changes either in nectar concentration ($F_{(1,43)} = 0.20$, $P = 0.658$; Fig. 2B) or in total sugar content per flower ($F_{(1,43)} = 2.09$, $P = 0.155$; Fig. 2C) throughout anthesis.

Step 3: comparison of nectar dynamics between unmanipulated flowers and flowers with successive nectar removals
The two sets of flowers under distinct experimental conditions had similar average nectar volume accumulated (Fig. 3A; $t_{(df = 36)} = -0.55$, $P = 0.58$). In contrast, the nectar concentration was lower in the set of flowers subjected to successive nectar removals (Fig. 3B; $t_{(df = 36)} = -4.\ 94$, $P < 0.001$). The total sugar produced per flower was quite similar between the two experimental sets of flowers (Fig. 3C; $t_{(df = 36)} = -1.40$, $P = 0.17$).

Micromorphology and histochemistry of the *Anemopaegma album* floral nectary

The floral nectary of *A. album* is a conspicuous and concentric globular structure that is slightly lobed, located at the base of the gynoecium, just below the ovary (Fig. 4A). Modified stomata with conspicuous pores were scattered throughout the nectary surface, and were slightly clustered in the upper region (Fig. 4B). A striated cuticle covered the nectary surface; single or clustered pores occurred primarily in the juxtaposition of radial walls (Fig. 4C). Accumulations of flocculent material, presumably dried nectar, were observed around the stomata (Fig. 4D) and on the nectary surface at pre-anthesis stage (Fig. 4E).

The floral nectary consisted of three distinct regions: uniseriate epidermis, nectary parenchyma and subnectary parenchyma *sensu* Nepi (2007). Histochemical examination indicated the presence of starch grains in all nectary cells, which decreased in size and abundance from the subnectary parenchyma towards the epidermis. In addition, polysaccharides were detected in the subcuticular space and vacuoles; pectin accumulations were identified in cell walls of epidermis and subjacent parenchyma, and in intercellular spaces of secretory parenchyma. All these substances were recorded in pre-anthesis buds and functional flowers. Oil droplets and phenolic substances were detected only in the parenchyma of secretory cells of functional flowers.

Cellular ultrastructure of the *Anemopaegma album* floral nectary

Analysis with TEM of the nectary in pre-anthesis buds showed epidermal and parenchyma cells compactly arranged, exhibiting a swollen middle lamella with variable electron density, suggesting cell wall degradation (Fig. 5A). The outer tangential walls of epidermal cells were thicker and covered with a continuous cuticle (Fig. 5B and C). Subcuticular spaces filled with flocculent material were observed (Fig. 5A). In this floral stage, epidermal cells had large nuclei, dense cytoplasm and well-developed vacuolar system (Fig. 5A and B). Abundant vesicles, Golgi bodies, endoplasmic reticulum segments (Fig. 5C and D) and amyloplasts (Fig. 5B and G) were the principal organelles in the cytoplasm. The secretory vesicles were of variable size and contained osmiophilic granules (Fig. 5D) or flocculent

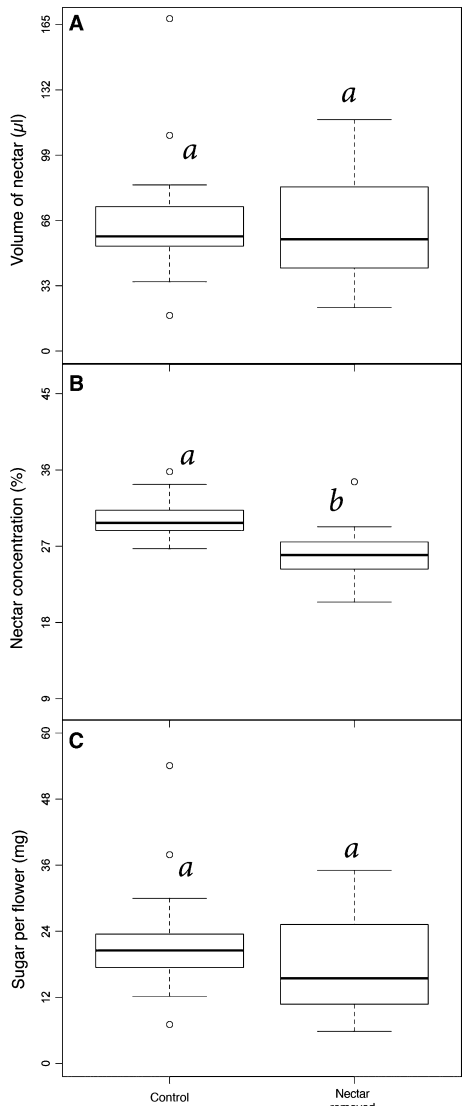


Fig. 3. Comparison between nectar accumulated in a set of unmanipulated flowers (n = 18 flowers, from 12 plants) and a set of flowers with successive nectar removals (n = 36 flowers, from 20 plants). Flowers of the two treatments were compared 24 h after the onset of anthesis. The nectar removal treatment was applied three times, at 0 h, 12 h and 24 h of anthesis, and the resulting nectar was used for subsequent analyses. Different letters within each graph represent significant differences between treatments (box plot with median and quartiles). A: Nectar volume (µl). B: Nectar concentration (percentage sucrose equivalents). C: The total sugar produced per flower (mg). For details, see Appendix S1.

material (Fig. 5E), and were commonly clustered adjacent to the vacuolar membrane (Fig. 5D) or fusing with the plasma membrane (Fig. 5E). The plasma membrane had a sinuous outline, showing in-foldings between the cell wall and protoplast (Fig. 5C and D). The RER (rough endoplasmic reticulum) profiles were irregular in size and generally situated adjacent to the mitochondria and plasma membrane (Fig. 5D and F). The plastids were devoid of lamellae, and contained small starch grains and granular electron-dense bodies, most likely proteins (Fig. 5G). Small vacuoles were distributed throughout the protoplasm, and a large vacuole was located at the distal pole of the cell (Fig. 5A and B); all vacuoles were filled with flocculent material, and may have also contained membrane debris (Fig. 5C). Plasmodesmata connected epidermal and parenchyma cells (Fig. 5B and F).

The nectary parenchyma cells of pre-anthesis buds had denser cytoplasm compared with epidermal cells (Figs 5A and 6A), and conspicuous nucleus and vacuoles of different sizes, irregularly distributed throughout the protoplasm (Fig. 6A). The cytoplasm was characterised by scattered vesicles containing electron-translucent or osmiophilic granules (Fig. 6B–E) and plastids (Fig. 6A). A high number of mitochondria with well-developed cristae were observed in clusters near the endoplasmic reticulum (Fig. 6C–E). The RER profiles were predominant in these cells, exhibiting irregular sizes and shapes and dilated regions (Fig. 6C–E). Near these swollen edges, there were many vesicles coated with ribosomes; their location and arrangement suggested that they had budded off from the RER cisternae (Fig. 6C). The amyloplasts here were larger and more irregular in shape (Fig. 6A) than in the epidermis (Fig. 5A). Amyloplasts with residual starch grains and signs of envelope rupture were common in these cells (Fig. 6F and G). Amyloplast debris was incorporated into the cytoplasmic matrix, and regions that were previously occupied by amyloplasts can be recognised by the abundance of flocculent material and presence of residual starch grains (Fig. 6H). When the plastid envelope also disintegrated, a new membrane was formed around these areas, forming new vacuoles that progressively fused together (Fig. 6G) to form one large vacuole (Fig. 6H).

The subnectary parenchyma region was characterised by intercellular spaces, and cells had a reduced cytoplasmic matrix distributed around the plastids and vacuoles (Fig. 6I). The plastids were larger than plastids of parenchyma cells, spherical in shape and filled with packed and conspicuous starch grains (Fig. 6I–K). The phloem was of sieve tube elements with prominent companion and parenchyma cells (Fig. 6I). Juxtaposition of RER cisternae with plasma membrane and/or plasmodesmata was commonly observed (Fig. 6J). Plasmodesmata connected subnectary parenchyma cells to one another and with phloem cells (Fig. 6I and J).

In functional flowers, epidermal cells were very similar to the previous stage, only differing in relation to development of the vacuome, which was bigger in the last stage. In addition, the previous distinction between nectary parenchyma and subnectary parenchyma was less clear in functional flowers. The parenchyma region was characterised by more developed intercellular spaces (Fig. 7A), some of which had accumulations of amorphous electron-lucent material (Fig. 7B). The parenchyma cells were characterised by a prominent nucleus, reduced cytoplasm and larger vacuoles in comparison with the previous stage (Fig. 7A). The population of membranous organelles was poorly developed, with plastids and mitochondria being more visible (Fig. 7C). The most striking difference from the previous stage was the change in size and content of plastids. At this stage, there was co-existence of different types of plastids: spherical plastids filled with starch grains (Fig. 7C) and ellipsoid plastids having thylakoids, small ellipsoid starch grains and electron-lucent oil droplets (Fig. 7D and E). The last type of plastid was predominant in functional flowers. Oil droplets were observed leaving the plastids towards the plasma membrane and traversing the tonoplast towards the vacuole (Fig. 7D). Osmiophilic material deposits were observed on the inner surface of the vacuole membrane juxtaposed to the plastid (Fig. 7D and E).

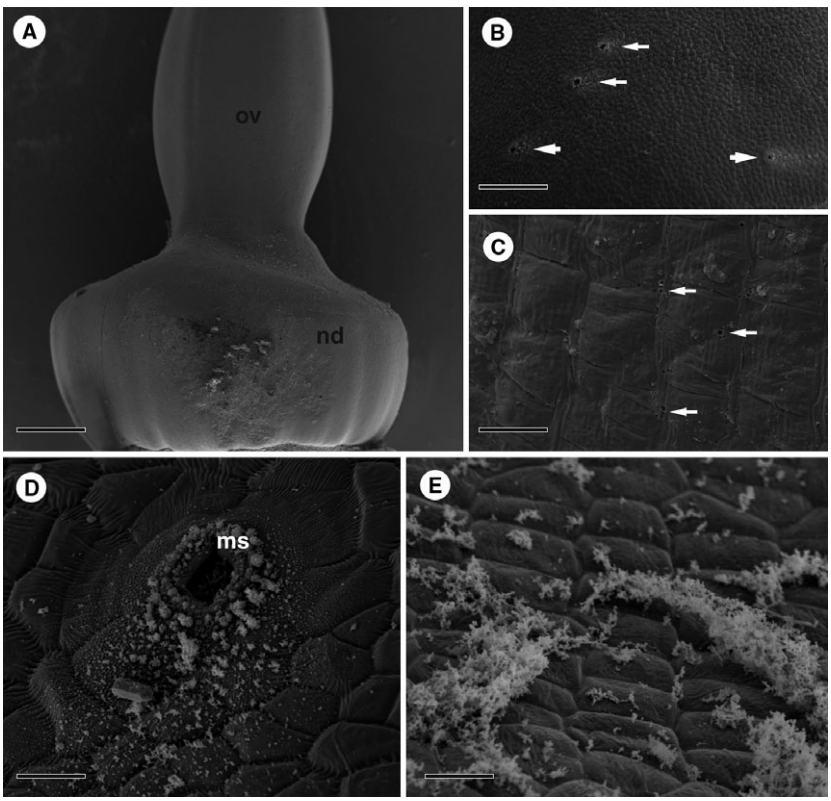


Fig. 4. The floral nectary of *Anemopaegma album* at the pre-anthesis stage (SEM). A: The nectary disc below the ovary. B: Detail showing raised stomata (arrows) at the upper region of the nectary. C: A nectary surface covered with the wrinkled cuticle; note cuticle pores (arrows). D: Flocculent material around the modified stomata and inside the pore. F: Flocculent material spread over the whole nectary surface. Scale bar: A: 500 μm, B: 300 μm, C: 30 μm, D, E: 20 μm. *Abbreviations*: ms, modified stomata; nd, nectary disc; ov, ovary.

DISCUSSION

The present study highlights the close association between nectary features and nectar production, which sustain interactions with animal pollinators that utilise nectar from flowers as their main energy resource. Through an integrative approach involving the manipulative study of floral nectar secretion as well as anatomical and ultrastructural investigations, our results stress the direct association between nectar secretion pattern and functional changes in the floral nectary at the cellular level. More specifically, we recorded higher production of floral nectar during the first period of anthesis, which is in accordance with the subcellular functioning of the floral nectary of *A. album*. The nectary cells in pre-anthesis buds revealed early storage and successive degradation of starch grains, the presumed source of sugars, which allowed production of substantial amounts of nectar at the pre-anthesis stage and rapid replenishment during the first 24 h of anthesis. This period is the most important for *A. album* pollination because the flowers are slightly protandrous, with a peak of stigma receptivity on the second day (Guimarães *et al.* 2015). Consequently, the observed pattern of nectar secretion and flower development could be related to potential activity time of pollinators, underlining the adjustments in this interaction.

The pattern of floral nectar secretion throughout flower development

The model of floral nectar secretion in *A. album* showed a trend of increasing nectar volume and total sugar per flower in the first hours of anthesis, followed by a decrease in both descriptors. This pattern could occur due to evaporation or re-absorption of the nectar accumulated inside the nectar chamber, as described for other plant species (Galetto *et al.* 1994; Nepi & Stipczyńska 2008; Stahl *et al.* 2012). We also recorded a linear decrease in nectar concentration throughout anthesis in this set of flowers. This observed trend of reduction in nectar volume, concentration and total sugar per flower in unmanipulated flowers may represent the capacity of this species to recover energetically valuable sugars and water that were not used by floral visitors, as pointed out in Nepi & Stipczyńska (2008). We did not find a general effect of repeated nectar removal on total nectar sugar produced per flower during anthesis. These results corroborate those seen in other Bignoniaceae species, in which total sugar production was not affected by repeated nectar removal, even though there was a decrease in concentration (Galetto *et al.*1994).

According to Cruden *et al.* (1983), the nectar secretion period is associated with the activity time of pollinators, and the start of secretion is adjusted such that enough nectar is available when pollinators first visit. Most *A. album* flowers fit this general pattern, producing nectar in pre-anthesis buds and providing a high volume of nectar when the flowers open, which also corresponds to the pattern described for other Bignoniaceae (Galetto *et al.* 1994). However, a set of *A. album* flowers showed no nectar at the moment of flower opening, with little nectar production in the subsequent floral stages. The nectar volume was more variable in repeatedly sampled flowers as compared to single sampled ones, which means that different flowers of *A. album* have different capacities for nectar replenishment. This variation in floral nectar availability could determine, at least in part, pollen dispersal among plants (Ott *et al.* 1985). For example, some plant species have flowers with

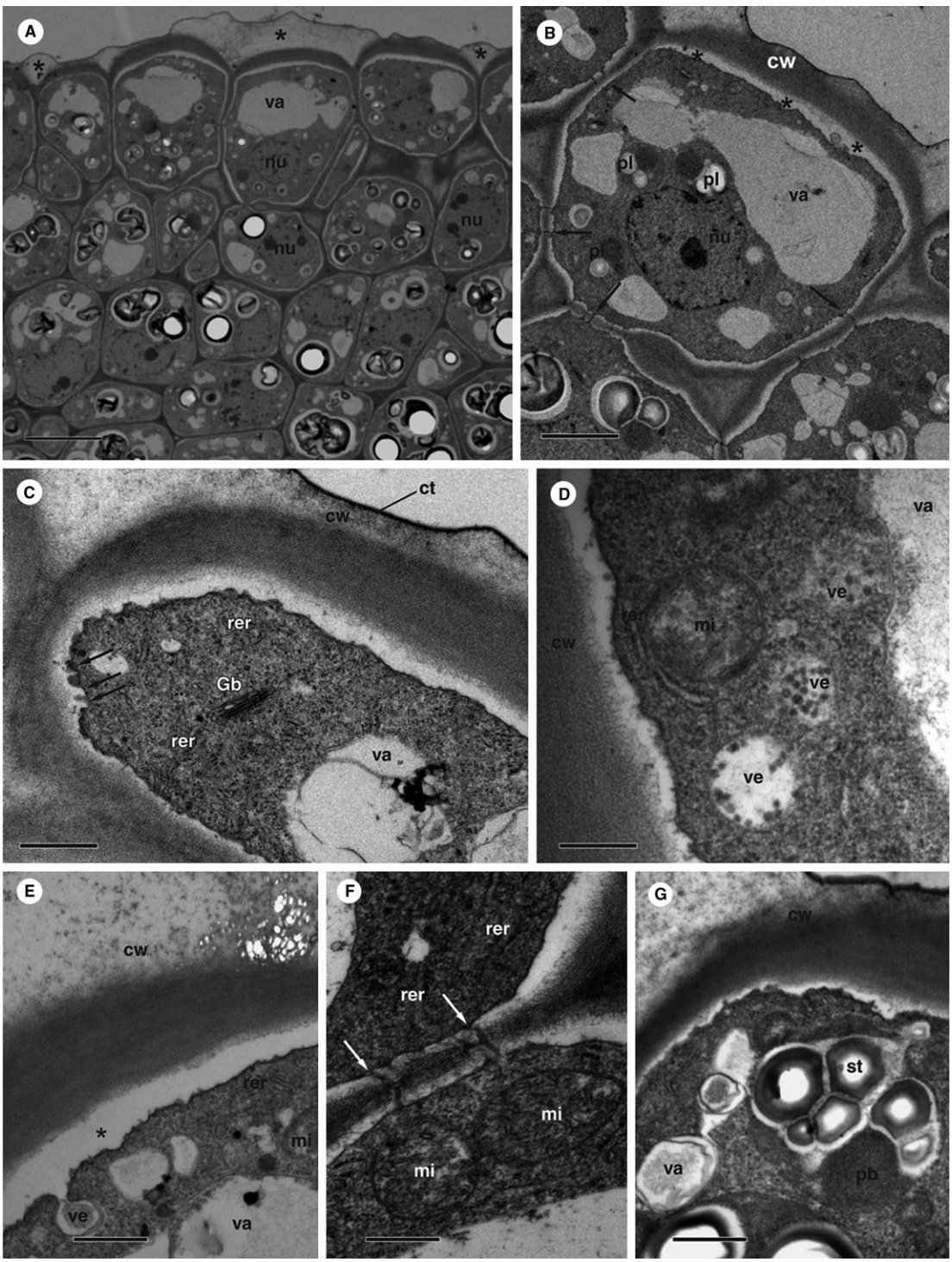


Fig. 5. Epidermal cells of an *Anemopaegma album* floral nectary at the pre-anthesis stage (TEM). A: A general view of the secretory epidermis. B: Plasmodesmata (arrows) connecting epidermal cells with each other and with subepidermal parenchyma cells. A larger vacuole in the apical cell pole containing flocculent material. Note the cell wall with loose fibrillar material (*). C: An epidermal cell showing electron-dense cuticle and abundant cytoplasm with ribosomes, rough endoplasmic reticulum and Golgi bodies. The vacuoles contain flocculent material and membrane debris. Note the vesicles (arrows) near the plasma membrane. D: Detail of peripheral cytoplasm from epidermal cells with free ribosomes, polyribosomes, mitochondria, rough endoplasmic reticulum and vesicles containing black inclusions. E: Note that the vesicle is filled with flocculent material fused with the plasma membrane. (F) Part of two epidermal cells showing mitochondria and RER cisternae close to plasmodesmata (arrows) in the anticlinal walls. G: A plastid with starch grains and a protein body (arrow). Scale bar: A: 5 μm, B: 2 μm, C: 1 μm, D, F: 0.5 μm, E, G: 0.7 μm. *Abbreviations*: ct, cuticle; cw, cell wall; Gb, Golgi body; mi, mitochondria; nu, nucleus; pb, protein body; pl, plastid; rer, rough endoplasmic reticulum; st, starch grains; va, vacuole; ve, vesicles.

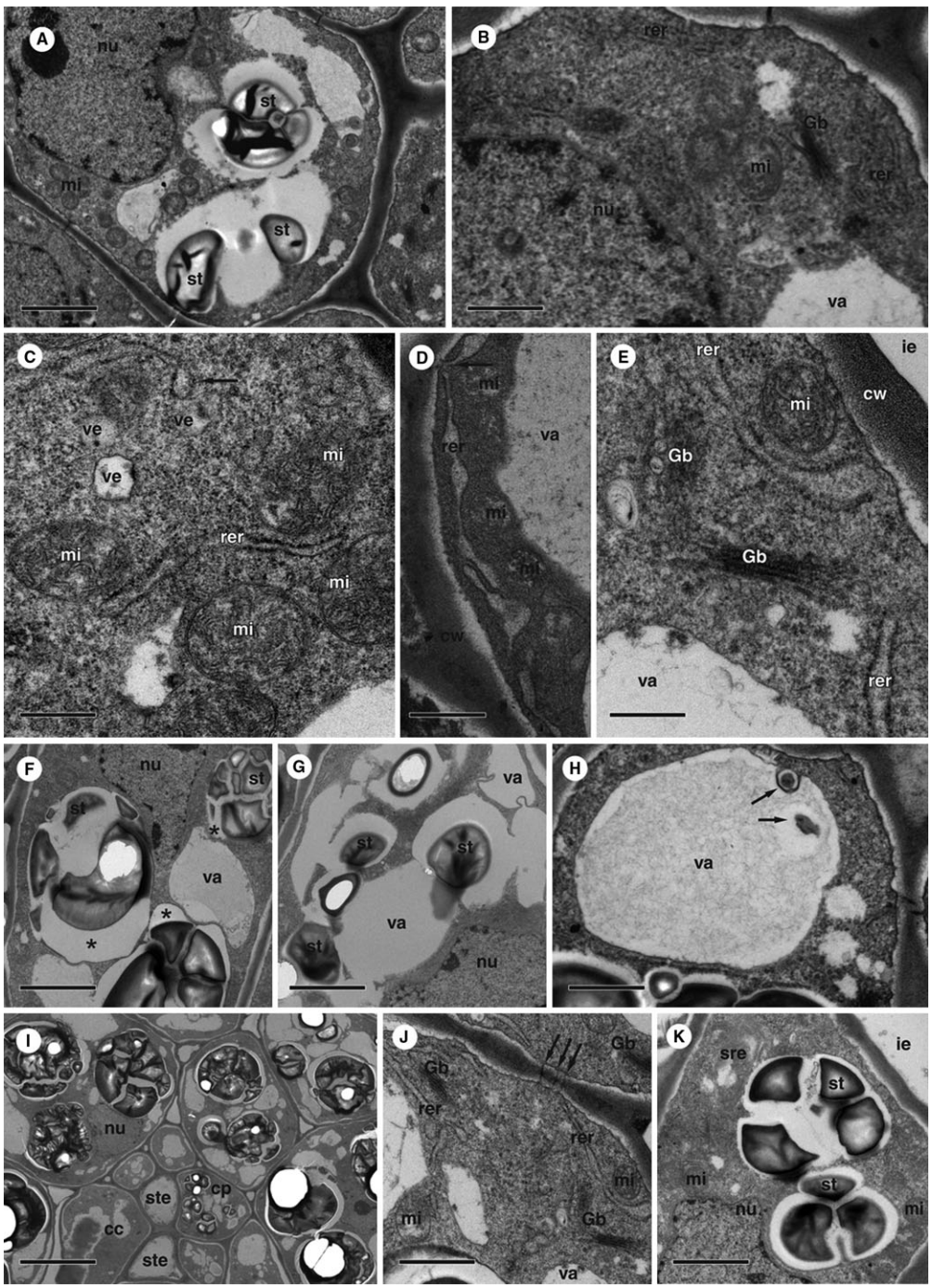


Fig. 6. Parenchyma cells of an *Anemopaegma album* floral nectary at the pre-anthesis stage (TEM). A–H: Nectary parenchyma; I–K: Subnectary parenchyma). A: A general view of nectar-secreting parenchyma cell. B: Cytoplasm with free ribosomes, polyribosomes, mitochondria, rough endoplasmic reticulum and vesicles. C: A vesicle coated with ribosomes (arrow), most likely derived from RER cistern vesicles containing osmiophilic inclusions, and globular mitochondria with well-developed cristae. D: Extensive endoplasmic reticulum with dilated regions in the cortical cytoplasm. Note the RER juxtaposed with plasmodesmata (arrow). E: Actively budding Golgi bodies with adjacent vesicles, globular mitochondria with well-developed cristae, and RER with dilated cisternae. F, G: Plastids with signs of starch grain degradation. F: The formation of lacunae (*) next to the starch grains within the plastids. G: The incorporation of plastids into the cytoplasmic matrix and the fusion of vacuoles. H: A new vacuole filled with flocculent material and starch grain debris (arrows). I: Sub-nectary parenchyma cells with very large amyloplasts. Note the sieve tube elements in association with conspicuous companion and parenchyma cells; the latter contain small amyloplasts with starch grains. J: A juxtaposition of RER cisternae with a plasma membrane and plasmodesmata (arrows) and active Golgi bodies. K: Plastids with signs of starch grain degradation. Scale bar: A, K: 1 μm, B: J 0.7 μm, C, E: 0.5 μm, D, H: 0.8 μm, F, G: 1 μm, I: 5 μm. *Abbreviations*: cc, companion cells; cp, parenchyma cell; cw, cell wall; Gb, Golgi body; ie, intercellular space; mi, mitochondria; nu, nucleus; rer, rough endoplasmic reticulum; sre, smooth endoplasmic reticulum; st, starch grains; ste, sieve tube element; va, vacuole; ve, vesicles.

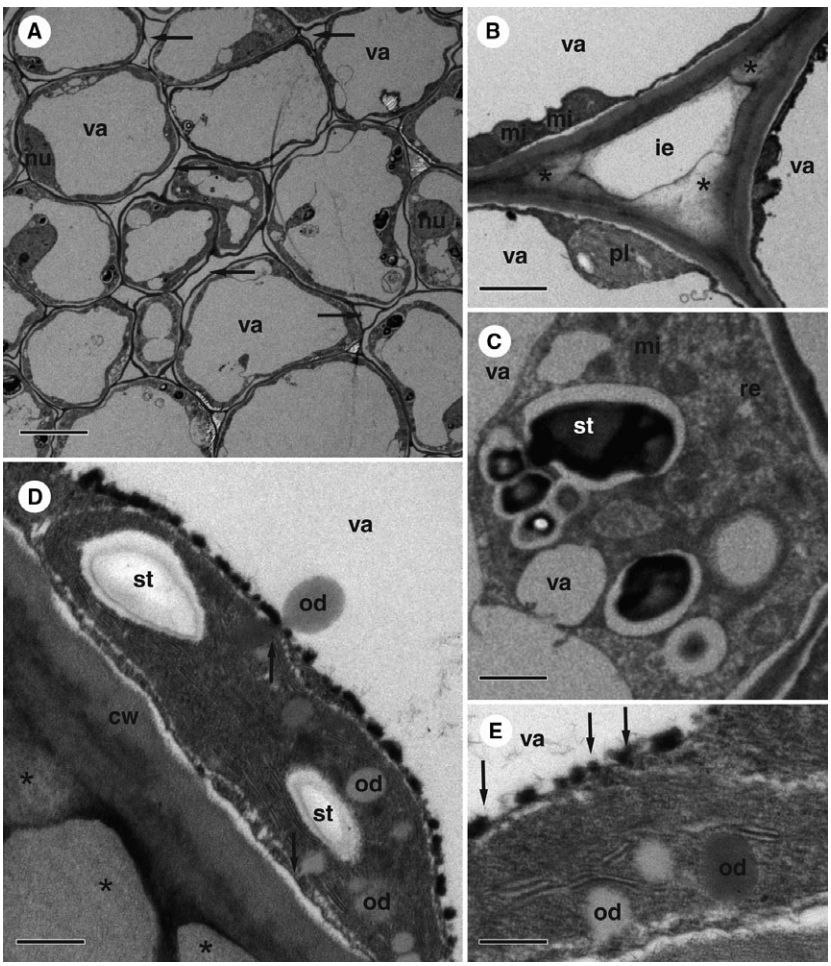


Fig. 7. Parenchyma cells of *Anemopaegma album* floral nectary of functional flowers (TEM). A: A general view of parenchyma cells with reduced cytoplasm and large central vacuole. Note the well-developed intercellular spaces (arrows). B: Accumulation of flocculent material (*) in the intercellular space. C: Peripheral cytoplasm with mitochondria, plastids, endoplasmic reticulum and Golgi bodies. D: An ellipsoid plastid with barely developed thylakoids, oil drops and small starch grains. Note the oil drops (arrows) leaving the plastids towards the plasma membrane and traversing the tonoplast toward the vacuole. The asterisk indicates accumulation of pectins. E: Detail showing oil inclusions inside the plastid and deposits of osmiophilic material at the inner surface of the tonoplast (arrows). Scale bar: A: 5 μm, B, C: 1 μm, D: 0.4 μm, E: 0.2 μm. *Abbreviations:* cw, cell wall; er, endoplasmic reticulum; mi, mitochondria; nu, nucleus; od, oil droplet; pl, plastids; st, starch grains; va, vacuole.

a large amount of nectar and flowers with no nectar in a bonanza–blank reward system (Brink 1982). Moreover, some blossoms may contain more than twice the nectar volume to the mean value of the population (lucky hits), which, on the one hand might represent an energy saving for some plants while on the other hand ensuring sufficient energy for foragers (Southwick *et al.* 1981). Both patterns that appear to occur in *A. album* could cause pollinators to forage more distant plants due to the higher variability in nectar resources, increasing the neighbourhood size and pollen dispersal distance (Ott *et al.* 1985). This can be especially significant in plant species that produce a large number of flowers at the same time (Anand *et al.* 2007), as observed in our focal species *A. album*.

In addition to volume variations, changes in nectar concentration may also occur during flower development, with nectar concentration typically increasing during anthesis, and the volume decreasing as a consequence of evaporation (Cruden *et al.* 1983). However, *A. album* flowers showed constancy in nectar concentrations throughout anthesis, despite high local temperatures and low relative humidity, especially at noon. Furthermore, the variability of nectar concentration was lower than that of volume and was similar between unmanipulated and manipulated flowers. The maintenance of constant concentration may be explained by active nectar regulation of the sugar content, presenting a balance among secretion, evaporation and re-absorption (Nepi & Stipczyńska 2008). Another possibility is that the mild viscosity of *A. album* floral nectar, which could be attributed to the presence of pectin from the wall of nectary epidermal cells, may represent an additional mechanism to reduce evaporation rates, particularly during drier hours (*e.g.* midday). Indeed, pectin is a strong hydrating compound that protects organs/structures from desiccation (Whistler & Smart 1953).

The lower nectar concentration observed in *A. album* flowers subjected to periodic nectar removal during the first 24 h of anthesis, relative to flowers that were measured only once, could be explained by exhaustion of carbohydrates stored in the nectary, as noted by Ren *et al.* (2007) in ornamental tobacco. Direct transport of sugars from the phloem has been suggested as the original mechanism for floral nectar production; however, starch accumulation in the nectary might be a derived strategy that could allow secretion of large amounts of sugar during the peak of secretion (Heil 2011 and references therein). Thus, the peak of nectary activity in *A. album* at pre-anthesis stage appears to be related to the metabolism of starch grains stored in nectary cells, and probably the direct contribution of phloem sugars is minor in the nectar secretion process. Another explanation could be the shorter time that nectar was exposed to the external environment, which could prevent evaporation of water in repeatedly sampled flowers, resulting in less concentrated nectar when compared to nectar from unmanipulated flowers.

Structural and ultrastructural features of the nectary during flower development

The floral nectary of *A. album* is an annular disc located below the ovary, as described for other *Anemopaegma* species (Rivera 2000). The release of nectar on the surface of the *A. album* floral nectary occurred in two ways: *via* modified stomata and through pores in the cuticle. The presence of modified stomata reinforces the idea that this is an important route of nectar release in species that secrete nectar within a short time period, as already described for other species with discoid floral nectaries (Davis & Gunning 1992; Fahn & Shimony 2001). The presence of pores in the cuticle, which were most frequently observed between adjoining epidermal cells at the pre-anthesis stage, at the peak of secretion, indicates that nectar passes along the apoplast and accumulates in the subcuticular space, as reported by Nepi (2007) for other plant species. The pressure of pre-nectar under the cuticle most probably causes its stretching, facilitating nectar release through the pores.

Remarkably, the main cellular changes in the nectary occurred at the pre-anthesis stage, coinciding with maximum secretion. The differences found in the ultrastructural organisation between the epidermis and nectary parenchyma of *A. album* suggests higher metabolic activity rates in the parenchyma region. When considering nectar liberation from the protoplast of epidermal cells, the presence of numerous mitochondria and secretory vesicles in the peripheral cytoplasm that fuse with the plasma membrane provide evidence of a granulocrine mechanism (Fahn 1979). On the other hand, low densities of Golgi bodies and of endoplasmic reticulum cisternae suggest the occurrence of an eccrine mechanism, in which molecules cross the plasma membrane by active transport (Nepi 2007; Vassilyev 2010). An eccrine mechanism is common when secretion is fast and copious and storage starch granules are involved (Pacini & Nepi 2007), as is the case of *A. album* flowers. The ultrastructural organisation of the nectary cells of *A. album* suggests that both mechanisms, eccrine and granulocrine, occur simultaneously.

Moreover, the accumulation of starch in plastids during the period preceding anthesis, as observed in the parenchyma cells of the *A. album* floral nectary, is characteristic of floral nectaries that produce abundant nectar (Nepi 2007; Ren *et al.* 2007). The high abundance of mitochondria and the juxtaposition of mitochondria with plastids in parenchyma cells may be related to energy requirements during starch hydrolysis (Fahn 1979). Our data provide strong evidence that this plastid starch storage is the principal source of carbohydrates of floral nectar, and that starches are hydrolysed to sugar at the secretion stage, as reported for other plant species (*e.g.* Fahn & Shimony 2001; Nepi 2007).

Furthermore, the conversion of amyloplasts in vacuoles was clearly observed in our study, and the region previously occupied by plastids remained well defined, suggesting that amyloplasts and vacuoles are the centres of nectar production, as reported in Peng *et al.* (2004). Plastid engulfment by vacuoles (Guiboileau & Masclaux-Daubresse 2012), which is likely an autophagic process, appears to be a common event in nectariferous tissues. RER dominated when compared with SER in parenchyma cells, which could be associated with autophagic processes related to the synthesis of proteins and the transport of hydrolytic enzymes that require a large amount of energy (Fahn 1979).

In the present study, the presence of ribosome-coated vesicles containing dense globules (most likely polypeptides) close to and in contact with the plasma membrane or plasmodesmata suggests intense exchange of secretion products inside and outside the protoplasts of nectary cells. The fusion of the endoplasmic reticulum to the *A. album* cell membrane and vesicles budding from the swollen edges of the ER cisternae in nectary parenchyma cells can be related to release into the apoplast (Fahn 1979). The floral nectary of *A. album* was supplied only by phloem, and it is probable that pre-nectar from the sieve tubes moves away from an apoplastic route *via* intercellular spaces and from cell walls, reaching the pores of modified stomata or the cuticle pores (*e.g.* Wist & Davis 2006). In *A. album*, the apoplastic transport of pre-nectar could be facilitated by the accumulation of pectin observed in the intercellular spaces of nectary parenchyma tissue. In addition, a symplastic pathway of pre-nectar could also occur in the floral nectary *via* the plasmodesmata that connect phloem elements with other nectary cells (Wist & Davis 2006; Vassilyev 2010).

Although amyloplasts are normally considered a final stage of plastid differentiation (Wheatley 1977), there was clear evidence of a conversion of the remaining amyloplasts to elaioplasts. The plastic nature of this organelle has been reported for floral nectaries of different species (Horner *et al.* 2007), particularly for nectaries that remain attached to the ovary base after nectar secretion has ended, as appears to be the case for *A. album*. The presence of oil droplets in the stroma of plastids, inside vacuoles or dispersed in the cytoplasm of the floral nectary is ultrastructural evidence of oil production during *A. album* anthesis. Although we have no data on the nectar chemical composition, oil produced in the floral nectary of *A. album* can participate in pollinator attraction or in deterrence of florivores (Raguso 2004). This deterrence of florivores could also be promoted by the presence of phenolic compounds, which were found in vacuoles of the nectary parenchyma cells of functional flowers of *A. album*. These substances tend to be water-soluble and occur in combination with sugars as glycosides, repelling certain visitors such as herbivores that feed on flower tissues (Harborne 1993), and therefore can contribute to the reproductive stages after pollination.

CONCLUDING REMARKS

Our study shows the complementarity of approaches involving manipulative studies of floral nectar production and investigations on the floral nectary at the cellular level, shedding light on nectar-mediated interactions. Although nectar removal from *A. album* flowers during the first 24 h of anthesis did not exert any influence on the pattern of floral nectar volume or total sugar, it did affect the nectar concentration. Starch accumulation in the nectary may allow the secretion of large amounts of sugar during the peak of secretion in *A. album* flowers. Indeed, we speculate that starch reserves are the principal source of floral nectar once the nectar concentration decreases after successive removals. Thus, the ultrastructural changes of nectary cells during flower development may lead to variations in nectar features, depending on the intensity of nectar collection by floral visitors.

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SUPPORTING INFORMATION

Additional Supporting Information may be found in the online version of this article:

Appendix S1. Diagram showing the experimental procedures carried out to characterise floral nectar secretion dynamics.

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